

# Koorosh Roohi

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## SUMMARY

Ph.D. Candidate in Biomedical Engineering at the University of Toronto and Faculty Affiliate Researcher at the Vector Institute, working across two research tracks: **3D point cloud processing from mmWave radar** and multi-modal sensing (RGB-D, IMU) for real-time healthcare activity recognition and infection risk prediction, and training-free diffusion models for high-resolution generative image synthesis. AI Manager at Masterly Inc., building **agentic AI systems**, including an **MCP host with maintainable MCP servers** powered by Vision-Language Models (VLMs) and Large Language Models (LLMs).

## EDUCATION

- **University of Toronto** Toronto, Canada  
*Ph.D. in Biomedical Engineering (Supervisor: Prof. Roshan Fekr and Mihailidis)* Sep 2023 - Nov 2028 (expected)
  - **GPA:** 4/4 (A+)
  - **Dissertation Title:** Infection Risk Estimation via Multi-modal Sensing and Deep Learning in Healthcare
- **Amirkabir University of Technology, (Tehran Polytechnic)** Tehran, Iran  
*B.Sc. in Computer Engineering* Sep 2018 - Mar 2023
  - **GPA:** 3.85/4  $\approx$  18.26/20

## RESEARCH EXPERIENCE

- **Research Assistant at University of Toronto** Toronto, Canada  
*Under the supervision of Prof. Roshan Fekr and Mihailidis* Apr 2023 - Present
  - Developing novel deep learning models to **quantify infection exposure risk** in real-time healthcare settings using **indoor localization** and **activity recognition**.
  - Fusing multi-modal sensor data (**mmWave radar**, RGB-D, IMU, BLE) to build robust environmental and motion-capture datasets for downstream activity-recognition model training and validation.
  - Proposed a novel **multi-modal fusion architecture** that dynamically enhances **mmWave radar**-based activity recognition, achieving around **10%** higher accuracy than single-modality models.
- **Faculty Affiliate Researcher at Vector Institute** Toronto, Canada  
*Generative AI & Diffusion Models* Mar 2026 - Present
  - Conducting research on **training-free diffusion model** methods for high-resolution image generation.
  - First author on **PhotoQuilt**, a training-free framework for arbitrary-resolution photomosaic synthesis.
  - Co-author on **SEGA**, a training-free method for resolution extrapolation in diffusion transformers.
- **Research Assistant at Amirkabir University of Technology** Tehran, Iran  
*Under the supervision of Prof. Safabakhsh* Dec 2021 - Mar 2023
  - Developed a **gait-based person re-identification** system using deep CNNs and **Horizontal Pyramid Matching (HPM)** layers, achieving **86%** accuracy on the **OUMVLP-Pose** dataset.
  - Implemented **triplet loss** for robust feature learning and deployed **YOLOv5** with **DeepSort** for real-world, multi-object tracking.

## WORK EXPERIENCE

- **AI Manager at Masterly Inc.** Toronto, Canada  
*AI-powered software solutions* Nov 2023 - Present
  - Engineered **Masterly Earendel**, a **Vision-Language Model (VLM)**-based video analysis system that automatically detects critical events, generating real-time alerts to reduce manual review in an agentic structure.
  - Leading development of **Masterly FinSuite**, a **Large Language Model (LLM)**-driven financial portal to automate reporting and provide decision support.
  - Deployed and manage scalable, production-level AI infrastructure on **AWS** (SageMaker, Bedrock, Lambda, EC2).
  - Supervised a team of **5** developers, delivering **2** production-ready AI modules.
- **Software Engineer at Crouse Company's Research & Innovation Department** Tehran, Iran  
*Computer Vision Specialist (Part-time)* Mar 2022 - Jun 2022
  - Built an **OpenCV**-based computer vision pipeline for **defect detection in car displays**, improving quality inspection accuracy over manual methods.
  - Implemented **Isolation Forest** and **Tesseract OCR** for pixel-level anomaly detection, successfully identifying **5** defect categories.
- **Client Developer at Quiz of Kings Studio** Tehran, Iran  
*Unity Game Engine Developer (Part-time)* Nov 2019 - Mar 2022
  - Contributed to client-side development for **Quiz of Kings**, Iran's most popular mobile game with **100k+** active users.
  - Led the migration of legacy code from **Corona** to **Unity**, boosting performance and maintainability of the final product.

## PUBLICATIONS & PRESENTATIONS

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- [Pre-print] Roohi, K., Rajabi, J., Fleet, A., & Taati, B. (2026). "PhotoQuilt: Training-Free Arbitrary-Resolution Photomosaics via Bootstrapped Tiled Denoising" *arXiv*.
- [Pre-print] Rajabi, J., Shaban, K., Roohi, K., Lindell, D. B., & Taati, B. (2026). "SEGA: Spectral-Energy Guided Attention for Resolution Extrapolation in Diffusion Transformers." *arXiv*.
- [Published] Roohi, K., & Roshan Fekr, A. (2026). "Comparative Analysis of Deep Learning Models for Real-time Nursing Activity Recognition." *Machine Learning: Health (IOP)*.
- [Published] Roohi, K., & Roshan Fekr, A. (2025). "A Comparative Analysis of Indoor Localization Technologies: A Review of Literature." *Computer Networks (Elsevier)*.
- [Published] Roohi, K., Esmailzahi, A., & Ahmad, M. O. (2025). "FUVT: A Deep Few-shot Unsupervised Learning-based Video-to-Video Translation Scheme using Kalman Filtering and Relativistic GAN." *Signal, Image and Video Processing (Springer)*.
- [Oral Presentation] Roohi, K., & Roshan Fekr, A. (2025). "Using mmWave Radar and Deep Learning to Classify Caregiver Activities for Infection Prevention" *The Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*.
- [Under Review] Roohi, K., & Roshan Fekr, A. "A Real-Time Caregiver Activity Recognition with Potential for Infection Prevention in Hospitals." *Smart Health (Elsevier)*.

## HONORS, AWARDS & PROFESSIONAL SERVICE

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- EPIC Doctoral Award (\$10,000 CAD), Emerging and Pandemic Infections Consortium (EPIC) (Jan 2025).
- Peer Reviewer, *Circuits, Systems, and Signal Processing (CSSP)* journal (13 reviews completed).
- Ranked 193<sup>rd</sup> / 140,000 Applicants, Nationwide University Entrance Exam (Jul 2018).

## TECHNICAL SKILLS

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- **AI & Deep Learning:** PyTorch, TensorFlow, State Space Models (Mamba4D), Diffusion Transformers (DiT), Vision-Language Models (VLMs), Multi-Agent Systems (MCP), Spatio-Temporal Modeling
- **3D Vision & Sensing:** 3D Point Clouds, 3D Object & Scene Understanding, mmWave Radar Signal Processing, Sensor Fusion (IMU/BLE), Computer Vision (YOLO, DeepSORT, OpenCV)
- **MLOps & Infrastructure:** AWS (SageMaker, Bedrock, EC2), Docker, Git, Linux, Real-time Inference, REST APIs
- **Languages & Data Science:** Python, C#, .NET, SQL, Pandas, NumPy, scikit-learn, Advanced Signal Processing

## TEACHING EXPERIENCES

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- **Teaching Assistant (Tutorial TA), University of Toronto - Department of Computer Science:** System Programming (Winter 2025, Winter 2026), Microprocessor Systems (Winter 2024), Software Design (Fall 2023, Fall 2024, Fall 2025)
- **Head Teaching Assistant, Amirkabir University of Technology - Department of Computer Engineering:** Principles of Artificial Intelligence (Fall 2022, Spring 2022), Programming Languages (Fall 2022), Algorithm Design (Fall 2021), Advanced Programming (Spring 2021), Fundamentals of Programming Lab (Winter 2021)